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Predicting Consumer Behavior from Sociodemographic Profiles:

An Exploratory Data Analysis of the Insurance Company Benchmark (COIL 2000)

Problem Statement

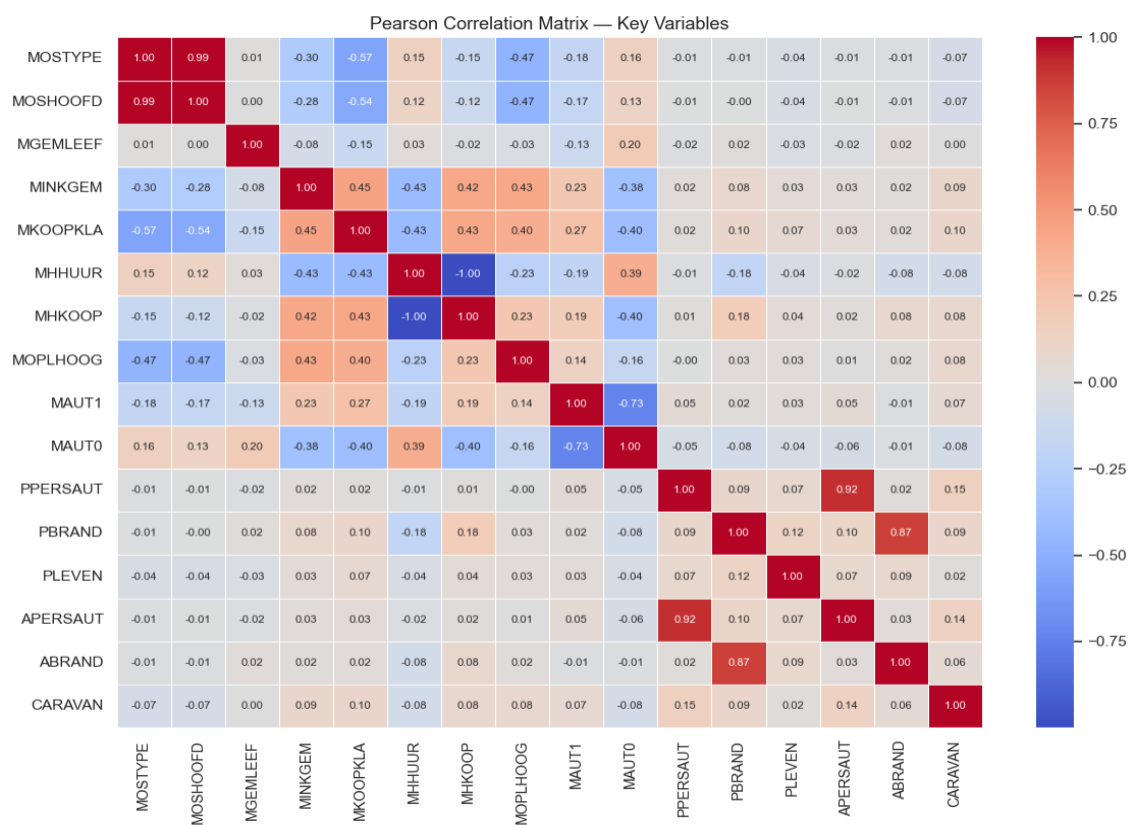
Who buys a mobile home policy? That is the central question this project set out to answer using the Insurance Company Benchmark (COIL 2000) dataset from the UCI Machine Learning Repository. The dataset contains 5,822 customer records from a Dutch insurance company, combining zip code level sociodemographic profiles with insurance product ownership data across 86 variables. The goal was to conduct a comprehensive exploratory data analysis to uncover which customer characteristics are associated with mobile home policy ownership, and whether those patterns could be used to identify likely buyers.

The Data Landscape

The first challenge was understanding what kind of data we were working with. The Pearson correlation matrix revealed that the dataset is effectively split into two worlds that barely communicate with each other. Sociodemographic variables like MINKGEM (average income), MKOOPKLA (purchasing power class), and MHHUUR (rented house %) are moderately to strongly correlated with each other, forming a coherent picture of neighborhood-level economic

life. Insurance product variables like PPERSAUT (car insurance contribution) and APERSAUT (number of car policies) are strongly correlated within their own group. Between the two groups, however, correlations are almost universally near zero.

This finding reframed the entire analysis. It meant that no single demographic variable would be a strong linear predictor of any individual product. The relationship between who a customer is and what they buy is not straightforward, it is a pattern that only emerges when customers are examined as a whole profile rather than variable by variable.

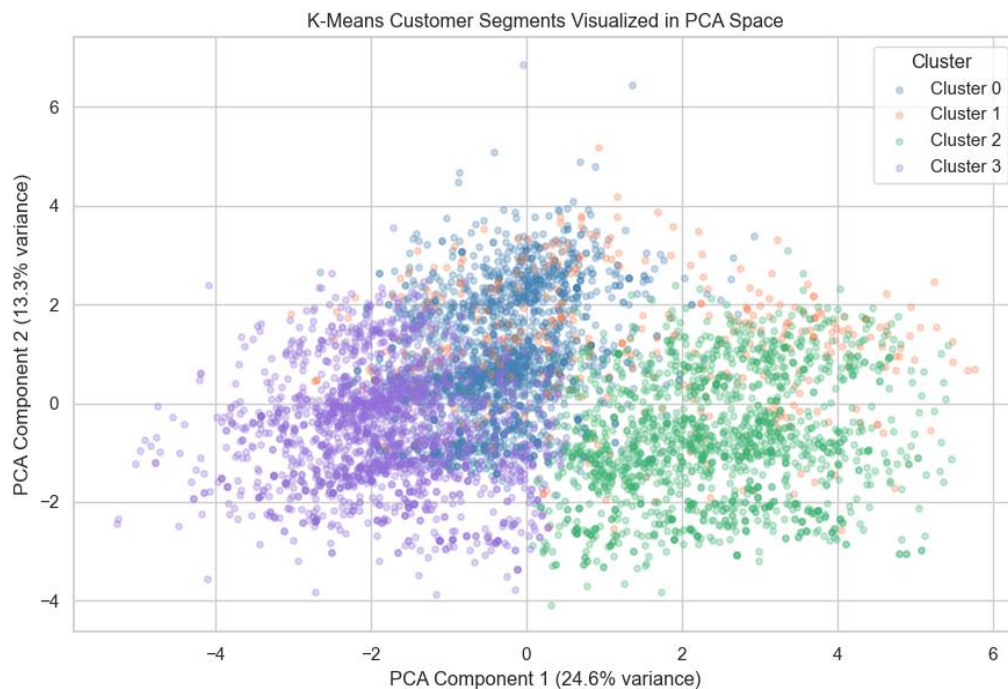


Finding the Signal Through Clustering

The K-Means clustering analysis confirmed this intuition. When customers were grouped based on their combined demographic and product profiles, four distinct segments emerged. The

PCA visualization showed these clusters occupying different regions of the reduced feature space, with Cluster 1 standing apart as a small but highly distinctive group of 344 customers.

Cluster 1 is the key finding of this project. Every single customer in this cluster owns a mobile home policy (CARAVAN = 1.000). They are characterized by the highest purchasing power (MKOOPKLA = 5.04), above average income (MINKGEM = 4.28), high car insurance contributions (PPERSAUT = 4.74), and moderate education levels. They are homeowners living in zip codes with relatively high car ownership rates. In contrast, Cluster 3, lower income renters with the lowest purchasing power, shows near zero mobile home policy ownership (CARAVAN = 0.002), and Cluster 2, the most affluent and educated segment, also shows zero ownership; suggesting that mobile home policies appeal to a specific middle-to-upper income bracket rather than the wealthiest customers.



Modeling Performance and Limitations

The hypothesis testing confirmed that income differences between CARAVAN owners and non-owners are statistically significant ($t = 6.994$, $p \approx 0$), and that car ownership rates in a customer's zip code are significantly associated with mobile home policy ownership ($\chi^2 = 36.99$, $p \approx 0$). However, translating these signals into a reliable predictive model proved difficult due to the heavy class imbalance, only 6% of customers own a mobile home policy.

Logistic regression outperformed Random Forest on this task, achieving a ROC-AUC of 0.757 compared to 0.671. The logistic regression correctly identified 72% of actual policy owners at the cost of a high false positive rate, which in a real marketing context is an acceptable tradeoff, it is better to contact 400 potential buyers and find 58 real ones than to miss them entirely. The Random Forest achieved higher overall accuracy (89%) but failed to capture the minority class meaningfully, a common pitfall when optimizing for accuracy on imbalanced data.

Recommendations

Based on the findings of this analysis, three actionable recommendations emerge for the insurance company. First, marketing campaigns for mobile home policies should prioritize customers who resemble Cluster 1, middle-to-upper income homeowners in zip codes with high car ownership rates and existing car insurance contributions. These customers represent the highest-density group of mobile home policy holders found in the data.

Second, sociodemographic data alone is insufficient for precise individual-level targeting. The near-zero correlations between demographic and product variables confirm that zip code profiles are too coarse to predict individual purchasing behavior on their own. Combining demographic profiles with existing product ownership history, particularly car insurance behavior, produces more meaningful signals as seen in the feature importance results.

Third, future modeling efforts should address the class imbalance more aggressively, potentially through oversampling techniques such as SMOTE, or by collecting additional behavioral features at the individual rather than zip code level. The current dataset's neighborhood-level demographics are a useful starting point but represent a fundamental ceiling on predictive precision.

Works Cited

- Van der Putten, Peter, and Maarten Van Someren. "CoIL Challenge 2000: The Insurance Company Case." Sentient Machine Research, 2000,
archive.ics.uci.edu/dataset/125/insurance+company+benchmark+coil+2000.
- UCI Machine Learning Repository. Insurance Company Benchmark (COIL 2000).
University of California, Irvine, 2000,
archive.ics.uci.edu/dataset/125/insurance+company+benchmark+coil+2000.